



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
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Alumni Relations and Career Development Office (ARCDO) Dashboard

Database System Implementation
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I. ABSTRACT

The Alumni Relations and Career Development Office (ARCDO) Dashboard is a web-based platform designed to improve data management for institutional records. The platform centralizes information related to Host Training Establishments (HTEs), Memorandum of Agreements (MOAs), On-the-Job Training (OJT) Coordinators, and Industry Partners (IPs).

Prior to this system, the ARCDO faced challenges in maintaining accurate and organized data using manual methods like spreadsheets and paper documentation. The new system addresses these issues by offering a secure and interactive solution with features such as an integrated dashboard, real-time data visualization, and efficient reporting tools.

The website ensures data security through role-based access control, where only Admins and Super Admins can modify or export data. Meanwhile, regular users are restricted to viewing relevant records. By combining modern web technologies with effective database management, the ARCDO Dashboard streamlines data handling, improves organizational efficiency, and supports better decision-making processes. This paper discusses the system's development, implementation, and potential future enhancements.



II. INTRODUCTION

Educational institutions manage extensive data related to student internships, industry partnerships, and institutional agreements. The Alumni Relations and Career Development Office (ARCDO) oversees this data, ensuring students are connected with established industry partners while tracking partnership agreements effectively.

Historically, the ARCDO relied on spreadsheets, printed documents, and manual entries to manage records. This approach resulted in various challenges such as inconsistent data, difficult access to key information, and time-consuming reporting processes. These inefficiencies hindered the office's ability to track MOA expiration dates, maintain accurate partnership records, and organize OJT coordinator details.

The ARCDO Dashboard was developed to solve these problems. By consolidating records into a structured database, the system offers improved data accessibility, faster record retrieval, and efficient reporting capabilities. The platform's intuitive dashboard provides visual insights through number cards, bar graphs, and donut charts, ensuring staff can easily assess institutional data.

Furthermore, the system's role-based access control ensures data security by assigning appropriate permissions to each user type. This ensures that only authorized personnel can modify records or export data, while regular users can securely access information without compromising its integrity. The ARCDO Dashboard represents a significant step toward modernizing data management for the institution.



III. REVIEW OF RELATED LITERATURE

The increasing demand for digital transformation in educational institutions has prompted the need for efficient alumni and career development office (ARCD O) management systems. Various studies have explored the benefits of digital record-keeping, data visualization, and secure database management in improving institutional workflows. Educational institutions worldwide are shifting towards digital platforms to enhance alumni engagement, streamline career services, and improve institutional operations (Pritchard & Wilson, 2019; Smith et al., 2021). The implementation of centralized database systems and web-based dashboards significantly contributes to more effective alumni tracking, career counseling, and employment support (Johnson, 2020). This section presents a comprehensive review of related literature supporting the development of the ARCD O Dashboard, focusing on digitalization, database systems, data visualization, and security.

Digitalization of Alumni and Career Services

The shift towards digital platforms in alumni and career services has been a growing trend in educational institutions worldwide. According to Pritchard & Wilson (2019), a centralized database system enhances alumni engagement by providing streamlined access to career resources, networking opportunities, and institutional records. Digital systems significantly improve operational efficiency, replacing traditional paper-based and spreadsheet methods, which are prone to errors and inefficiencies.

A study conducted by Smith et al. (2021) highlights the advantages of web-based alumni tracking systems, noting that such systems enhance institutional connectivity with former students. By utilizing digital platforms for job placement tracking and Memorandum of Agreement (MOA) monitoring, universities can maintain up-to-date partnerships with industry stakeholders.



Furthermore, digital career services provide a structured framework for career counseling and job matching, thereby increasing student employability (Johnson, 2020).

Additionally, studies such as those by Anderson & Roberts (2020) emphasize the importance of user-friendly interfaces in digital career service platforms. The ARCDO Dashboard aligns with these findings, aiming to modernize alumni and career services through a centralized, accessible, and secure digital platform. Furthermore, the integration of real-time updates ensures that both alumni and career offices maintain seamless communication, fostering a more engaged and connected alumni network. These digital solutions allow career services to be more responsive, efficient, and adaptable to the evolving demands of the job market.

Database Systems for Institutional Record Management

Effective data management is crucial for institutions handling vast amounts of information. Database systems provide structured and systematic approaches to storing, retrieving, and managing data (Gupta & Mehta, 2020). The use of relational databases, such as MySQL and PostgreSQL, has been widely adopted in academic institutions for handling alumni and career development records.

The study by Brown & Taylor (2022) discusses the importance of role-based access control (RBAC) in database security. Implementing RBAC ensures that only authorized personnel can modify or retrieve specific data sets, reducing the risk of unauthorized access. Similarly, Chandra & Kumar (2021) emphasize that database normalization techniques improve system efficiency by eliminating redundancy and ensuring data integrity.

The ARCDO Dashboard incorporates these security principles by restricting data modification and export privileges to Admins and Super Admins while allowing regular users to access relevant information securely. Additionally, the system employs structured query language (SQL)



optimizations to enhance query performance, ensuring quick retrieval of essential alumni records. These database systems play a pivotal role in maintaining the accuracy, consistency, and security of institutional records, ultimately enhancing the efficiency of alumni and career services.

The Role of Dashboards in Data Visualization

Dashboards are essential tools for presenting complex data in a user-friendly manner. According to Johnson & Lee (2021), interactive dashboards enhance decision-making by providing real-time insights through visual representations like bar graphs, pie charts, and heatmaps. The study emphasizes that well-designed dashboards help users quickly analyze trends, identify patterns, and make data-driven decisions.

Research by Anderson & Roberts (2020) highlights that data visualization significantly improves user engagement and comprehension, particularly in administrative decision-making. By incorporating user-centered design principles, dashboards become more intuitive and accessible to various stakeholders, including university administrators and career officers.

Furthermore, studies by Zhang & Li (2021) emphasize the importance of customizable dashboards, allowing institutions to tailor their visualizations based on specific operational needs. The ARCDO Dashboard adopts these principles by offering real-time data visualization tools such as number cards, bar graphs, and donut charts. These features enable ARCDO personnel to monitor MOA statuses, industry partnerships, and alumni records efficiently. Additionally, quick access tables and search functionalities contribute to a seamless user experience, facilitating efficient decision-making. The integration of visually appealing and easy-to-interpret data representations enhances the effectiveness of career service initiatives and alumni engagement strategies.



Security and Data Protection in Web-Based Systems

Data security is a critical aspect of any digital platform handling sensitive information. Research by Chen et al. (2020) highlights the importance of implementing encryption, authentication mechanisms, and regular data backups to safeguard institutional records. Security breaches in alumni databases can lead to unauthorized data access and identity theft, making robust security measures a necessity.

Further studies by Zhang & Li (2021) discuss the integration of multi-factor authentication (MFA) in web-based systems to enhance user security. Their research indicates that MFA reduces unauthorized access incidents by 70%, significantly improving data protection. Web-based platforms must adopt these security measures to ensure the confidentiality, integrity, and availability of stored data.

Additionally, Brown & Taylor (2022) highlight the role of regular security audits and penetration testing in maintaining secure digital infrastructures. The ARCDO Dashboard addresses these concerns through role-based access control, secure login authentication, and backup procedures. Features such as email verification for password recovery further enhance system security, ensuring that only authorized users gain access to sensitive data. Additionally, the dashboard employs Secure Sockets Layer (SSL) encryption to protect data transmissions between users and the server. By integrating these security measures, the ARCDO Dashboard provides a reliable and secure platform for managing alumni and career development data.



IV. SYSTEM DESIGN

The **ARCDO Dashboard** was designed with security, scalability, and usability in mind. Its structure follows a modular design that ensures organized development, simplified maintenance, and improved user experience.

Login and Authentication

The system implements secure user authentication to protect user accounts and ensure controlled access. Key features include:

- **Email and Password Authentication:** Users are required to log in using their registered email and password. Only verified accounts can access the system.
- **Forgot Password Feature:** Users who cannot remember their password can utilize the **Forgot Password** option to reset their credentials securely via email verification.
- **"Remember Me" Option:** For convenience, users can enable the **Remember Me** feature, which allows faster logins by securely storing their credentials in their browser.

Dashboard Features

The ARCDO Dashboard offers an organized, data-driven interface that displays key information efficiently. Core dashboard features include:

- **Number Cards Expansion:** Clicking on number cards expands detailed insights:
 - **HTEs, MOAs, and OJT Coordinators** — Expands to display detailed distribution data.
 - **Industry Partners (IPs)** — Expands to display a **Donut Graph** representing the status distribution of IPs.



- **Business Nature Bar Graph (NoB):** Displays the distribution of business types for HTEs and IPs. Users can hover over specific bars to reveal exact counts.
- **MOA Status Graph:** Displays a **Donut Graph** showing MOA statuses categorized as active, pending, or expired.
- **Quick Access Table (QAT):** Provides a swappable, year-filtered list of HTEs, IPs, and OJT Coordinators for quick navigation and reference.
- **Search Functionality:** Users can efficiently search HTE, IP, and OJT Coordinator data using keywords or filters for fast record retrieval.
- **Refresh Button:** Located on the upper right-most part of the screen, this feature updates the displayed content to ensure users see the most recent data.

User Management

The system integrates role-based access control to manage user permissions and ensure data security. Key features include:

- **User Roles:** The system defines three distinct roles — **Super Admin**, **Admin**, and **User** — to control access levels effectively.
- **Admin Access Management:** The **Super Admin** has exclusive control over assigning admin privileges via the **User Access** panel.
- **Account Management:** Users can manage their account details (e.g., contact number, password) within the **Account** settings.

Data Management

The system offers robust data management tools, available only to **Admins** and **Super Admins** to ensure data integrity. Key features include:



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- **Add New Entries:** Admins can create new records for HTEs, MOAs, IPs, and OJT Coordinators using structured data entry forms.
- **Edit Entries:** Existing records can be modified to reflect updated information.
- **Delete Entries:** Admins can permanently remove outdated or incorrect entries.
- **Export Data:** Admins can export data for HTEs, IPs, and OJT Coordinators in **CSV format** for reporting purposes.

Security & Data Protection

The ARCD0 Dashboard includes security features to protect sensitive data and user accounts.

- **JWT Authentication:** The system uses **JSON Web Tokens (JWT)** to securely manage user sessions. Tokens are stored safely to prevent unauthorized access.
- **Environment Variables (.env):** Sensitive credentials like database passwords, Azure Storage keys, and JWT secrets are securely stored in an .env file to prevent exposure.
- **Session Management:** Users are automatically logged out after prolonged inactivity to prevent unauthorized access.
- **Role-Based Access Control:** Ensures that only Admins and Super Admins can modify or export data, while Users are restricted to viewing content.

Data Storage and Backup

To ensure data consistency, reliability, and security, the system uses:

- **MySQL Database on Azure:** The system leverages **Azure Database for MySQL** to manage and store records securely.



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- **Azure Storage for MOA Documents:** Important MOA files are stored securely using **Azure Blob Storage**, ensuring documents are protected and accessible only to authorized personnel.
- **Backup Protocols:** Regular database backups ensure data recovery in the event of unexpected failures or corruption.

The ARCD0 Dashboard's design balances usability with strong security measures, ensuring users can efficiently manage data while protecting sensitive information. This modular structure allows for future scalability, ensuring the system can expand to meet evolving institutional needs.



V. IMPLEMENTATION DETAILS

USER MANUAL

User Manual Overview

The ARCDO Dashboard prioritizes user experience with clear navigation, intuitive design, and practical features to improve data handling:

- **Dashboard Overview:** Displays total counts for HTEs, MOAs, IPs, and OJT Coordinators. Clicking number cards expands data insights, showing distribution details or status graphs.
- **Search Functionality:** Users can search data entries for HTEs, IPs, and OJT Coordinators using flexible search filters for faster record retrieval.
- **Quick Access Table (QAT):** Provides a swappable, year-filtered table for HTE, IP, and OJT Coordinator data, enabling quick navigation.
- **Export Feature:** Admins and Super Admins can download data in **CSV format**, simplifying reporting.
- **Role Management:** The **Super Admin** can assign Admin roles via the **User Access** feature.



Introduction

The ARCDO Database Website is a comprehensive platform designed to simplify and enhance the management of various institutional data points. By centralizing records related to Host Training Establishments (HTEs), Memorandum of Agreements (MOAs), On-the-Job Training (OJT) Coordinators, and Industry Partners (IPs), the system ensures that essential information is always organized, accessible, and easy to manage. This manual offers detailed instructions to guide users through the platform's key features, ensuring they can efficiently navigate and manage records.

User Roles and Access Levels

The ARCDO Database Website assigns roles to ensure data security and maintain organized workflows:

- **Administrator/Super Administrator:** Administrators have full system control, allowing them to manage users, oversee data entries, and adjust system settings. Only administrators can export data for HTEs, IPs, and OJT Coordinators.
- **Staff Users:** Staff users have access to manage entries related to HTEs, MOAs, OJT Coordinators, and IPs. They can modify records and track MOA statuses but cannot export data.
- **Alumni/Students:** This role is restricted to read-only access. Alumni and students can explore HTEs, IPs, and MOA data relevant to their career development but cannot modify or export information.

Logging In

1. Navigate to the ARCDO Database Website via your web browser.
2. Enter your registered **username** and **password** in the respective fields.



3. Click the **Login** button to proceed. If your credentials are correct, you will be redirected to the main dashboard. If the login fails, use the **Forgot Password** link to reset your credentials securely.

Dashboard Overview

- **Refresh Page:** Located on the upper right-most part of the page, this button refreshes the current content to display the most up-to-date data.
- **Number Cards Expansion:** Clicking on the number cards will expand them to display detailed data:
 - **HTEs, MOAs, and OJT Coordinators:** Expands to show the distribution of numbers within their respective categories.
 - **Industry Partners (IPs):** Expands to display a donut graph illustrating IP statuses for better visual representation.
- **NoB Bar Graph:** Users can hover over specific bars in the Business Nature Graph to reveal exact counts for better clarity.
- **Quick Access Table (QAT):** Provides a swappable, year-filtered list of HTEs, IPs, and OJT Coordinators for quick navigation and reference.
- **Refresh Page:** Located on the upper right-most part of the page, this button refreshes the current content to display the most up-to-date data.
- **Number Cards Expansion:** Clicking on the number cards will expand them to display detailed data:
 - **HTEs, MOAs, and OJT Coordinators:** Expands to show the distribution of numbers within their respective categories.
 - **Industry Partners (IPs):** Expands to display a donut graph illustrating IP statuses for better visual representation.



- **Refresh Page:** Located on the upper right-most part of the page, this button refreshes the current content to display the most up-to-date data. The dashboard offers an organized, at-a-glance view of essential system data. Key features include:
- **Total Numbers:** Displays the total number of HTEs, MOAs, OJT Coordinators, and IPs currently stored in the database.
- **Business Nature Graph:** A visual bar graph representing the distribution of business types for both HTEs and IPs.
- **MOA Status Graph:** A donut graph illustrating the status of MOAs, categorized as active, pending, or expired.
- **Quick Access Table:** Provides a swappable, year-filtered list of HTEs and IPs for quick navigation and reference.

Managing Data

Note: Adding, editing, and deleting entries are restricted to **Administrators** and **Super Administrators** only.

Searching Entries

1. Navigate to the respective data section (HTEs, IPs, or OJT Coordinators).
2. Use the **Search Bar** at the top of the section.
3. Enter keywords such as organization name, contact person, or year.
4. Filter options may be available to refine your search results.
5. Results will automatically update as you type for quick access to relevant records.

Adding Entries

1. Navigate to the desired data section (HTEs, MOAs, etc.).
2. Click the **Add New** button to start data entry.
3. Fill in all relevant fields as required. Each form includes detailed prompts to



ensure accurate data entry.

4. Click **Save** to securely store the new entry in the system.

Editing Entries

1. Locate the entry you wish to modify using the search or filter options.
2. Click the **Edit** button next to the entry.
3. Update the necessary details.
4. Click **Save** to confirm your changes.

Deleting Entries

1. Locate the entry to be deleted.
2. Click the **Delete** button beside the record.
3. Confirm the deletion when prompted to permanently remove the data.

Exporting Data (Admin/Super Admin Only)

1. Navigate to the respective data section (HTEs, IPs, or OJT Coordinators).
2. Click the **Export** button available in each section.
3. The exported file will download automatically in CSV format.

Users without administrative privileges cannot access export functionality and are limited to viewing data via the dashboard only.



DEVELOPER MANUAL

Developer Manual Overview

The system's backend uses modular development practices to simplify code management and ensure scalability:

- **Database Management:** Tables are structured with proper indexing and foreign keys to ensure efficient data access and referential integrity.
- **API Development:** Using ExpressJS, endpoints were created to manage CRUD operations securely. Middleware functions ensure data validation, error handling, and secure user authentication.
- **Deployment Configuration:** Sensitive credentials, including database passwords, JWT keys, and Azure Storage details, are stored securely in a .env file to prevent unauthorized access.



Introduction

The Alumni Relations and Career Development Office (ARCDO) Database is a digital platform designed to streamline data management, enhance operational efficiency, and support career development initiatives. This developer manual serves as a comprehensive guide for understanding the technical implementation, architecture, and functionalities of the system.

The ARCDO Database is built with a modern web technology stack to ensure scalability, security, and seamless user experience. It manages data related to Host Training Establishments (HTEs), Industry Partners, Memorandum of Agreements (MOAs), and On-the-Job Training (OJT) coordinators. This document provides a structured overview of system architecture, database relationships, key features, deployment guidelines, and maintenance best practices to assist developers in managing, updating, and extending the system efficiently.

System Architecture

The system is designed with a robust architecture that integrates modern web technologies to ensure scalability, security, and efficient data handling.

- **Frontend:** Developed using **ReactJS** with **TailwindCSS** for dynamic and responsive user interactions. ReactJS's component-based structure ensures modular development, while TailwindCSS simplifies styling and ensures consistency across the platform.
- **Backend:** Powered by **NodeJS** with **ExpressJS** as the server-side framework. ExpressJS provides lightweight routing and middleware capabilities to manage requests, handle data processing, and ensure security.



- **Database:** Uses **MySQL** with **Azure Database for MySQL** for structured data storage. Azure Database ensures high availability, automated backups, and enhanced scalability for future data growth.

Database Structure

ARCDO Tables and Relationships

- `admin_profile`: Stores administrative user details and settings.
- `hte` (Host Training Establishments): Contains details of organizations providing training opportunities.
- `industry_partner`: Maintains records of companies collaborating with ARCD.
- `moa` (Memorandum of Agreements): Stores details of agreements between ARCD and external entities.
- `ojt_coordinator`: Contains information about coordinators managing on-the-job training programs.
- `user_account`: Manages authentication and authorization for different user roles.
- `users`: Stores general user details and profiles.

MOA Monitoring Tables

- `moa_documents`: Stores uploaded documents related to MOAs.
- `moa_info`: Maintains additional metadata and tracking details for MOAs.

Relationships

- Each `moa` record links to an `hte` and `industry_partner` through foreign keys.
- `moa_documents` and `moa_info` are directly related to `moa`, ensuring proper document



tracking.

- `user_account` and `admin_profile` are linked to users, enabling role-based authentication.

Key Features and Implementation

Authentication System

- Utilizes ExpressJS middleware for authentication with role-based access control (RBAC) to restrict data access based on user roles.
- Integrates Two-factor authentication (2FA) for administrator accounts to enhance security and prevent unauthorized access.

Dashboard Design

- Uses ReactJS components for dynamic data visualization with Chart.js for graphical insights such as bar graphs and donut charts.
- Implements AJAX requests to asynchronously fetch and update dashboard data without page reloads, improving performance.

Data Management

- Provides full CRUD (Create, Read, Update, Delete) functionality for managing HTEs, MOAs, OJT Coordinators, and IPs.
- Utilizes optimized SQL queries and indexing to enhance data retrieval speed.
- Implements user-friendly search and filter options for improved data accessibility.



Report Generation

- Reports are generated using dynamic ReactJS components to display data in both tabular and visual formats.
- Offers data filtering options based on date range, status, and category for tailored report creation.

Frontend Pages Implementation

The frontend includes various key pages designed with ReactJS and TailwindCSS:

- Add Data Page: Renders AddData and navigation components for adding new records.
- Error Page: Provides a custom 404 error page with a back-to-dashboard button.
- HTE Dashboard Page: Displays HTE-related data and navigation components.
- Industry Partners Page: Manages and displays industry partners' details.
- Login Page: Facilitates user authentication and entry to the platform.
- MOAs Page: Displays detailed information about Memorandum of Agreements.
- Overview Page: Offers a consolidated summary of system data with dynamic visualization components.

Deployment Process

.env Configuration for Deployment

```
PORT=3001

#database

DB_HOST=arcdo.mysql.database.azure.com
DB_USER=ArcdoAdmin
DB_PASS=dashboardDB1
DB_NAME=arcdo

#moa database
MOA_DB_HOST=pupmoamonitoring.mysql.database.azure.com
MOA_DB_USER=mms
```



```
MOA_DB_PASSWORD=PUP_moa_monitoring
MOA_DB_NAME=moamonitoring

JWT_SECRET=mySuperSecretKey123!
AZURE_STORAGE_ACCOUNT_NAME=pupmoafilesstorage
AZURE_STORAGE_ACCESS_KEY=iJVOyFLp541gAaT0AJvh+S/fKwo2J7QF1FzXl6xoCXi1VeN
  AQBki6+Ivmyd3DFGsoQQpaXyezYXW+AStWRqjnA==
AZURE_STORAGE_CONTAINER_NAME=moa-documents
AZURE_STORAGE_CONNECTION_STRING=DefaultEndpointsProtocol=https;AccountNa
  me=pupmoafilesstorage;AccountKey=iJVOyFLp541gAaT0AJvh+S/fKwo2J7QF1FzX
  l6xoCXi1VeNAQBki6+Ivmyd3DFGsoQQpaXyezYXW+AStWRqjnA==;EndpointSuffix=c
  ore.windows.net

#jwt
JWT_SECRET_KEY                                     =
  32406ac5060629433233c7c65f99a32d5b850a62273b7845332125ff8caa952ca6e6a
  13c345514f55e11ce2333eafc5ea63e4375d56672fbdc770f6f3f033196
JWT_REFRESH_SECRET_KEY                             =
  cd37aa858c1b70eb9724c53297e9d77f8b51cbfe18732fe02500030ad376142790402
  438511674e77a5663b6d9b2bb533f23e832dd5f419ed7687623ac08b31d
JWT_EXPIRES_IN = 15min
JWT_REFRESH_EXPIRES_IN = 7d

#nodemailer
EMAIL_USER=puparcdashboard@gmail.com
EMAIL_PASS=wver owtd jxka ynpt

# EMAIL_PASS=ejok zodh jwpj stnh
CLIENT_URL = http://localhost:5173
```

Deployment Instructions

- Clone the repository using **git clone**.
 - (<https://github.com/edelmode/arcdo/tree/main>)
- Install dependencies using **npm install**.
- Configure the **.env** file with database credentials, Azure environment details, and other required settings.
- Run database migrations using **npm run migrate** or relevant migration scripts.
- Start the development server with **npm run dev**.



Maintenance and Updates

- Apply security updates promptly by running **npm update**.
- Clear caches with **npm run cache:clear** to ensure data refreshes properly.
- Regularly monitor database performance using Azure Database Insights for query optimization and load balancing.

Error Handling and Logging

- Utilizes ExpressJS's error-handling middleware for structured error reporting and recovery.
- Implements logging tools such as Winston and Morgan to track system errors, access logs, and user activities.
- Logs are automatically stored in a secure Azure storage service for traceability and auditing.

Backup and Recovery

- Schedules daily database backups using Azure's Backup and Recovery Services to safeguard data integrity.
- Maintains proper version control using Git with feature-specific branches for easier troubleshooting and rollback.



VI. DISCUSSION

The ARCDO Dashboard represents a significant advancement in improving data management processes for the Alumni Relations and Career Development Office. By centralizing records for HTEs, MOAs, OJT Coordinators, and Industry Partners, the system addresses the inefficiencies commonly faced with manual documentation methods. The platform's key features — including its interactive dashboard, search functionality, and export tools — are designed to streamline workflows, reduce data retrieval time, and improve overall accuracy.

The implementation of the ARCDO Dashboard reflects the institution's commitment to enhancing operational efficiency and ensuring data security. The integration of role-based access control effectively limits data modification and export permissions to Admins and Super Admins, preventing unauthorized changes. Meanwhile, the User role offers a secure, read-only experience for staff and students seeking information. The dashboard's visual elements, such as the expanded number cards, business nature bar graph, and MOA status chart, provide clear insights that support informed decision-making.

The system's emphasis on security further strengthens its reliability. Features such as JWT Authentication, session management, and secure environment configurations ensure that sensitive information remains protected. Additionally, the use of Azure Database for MySQL and Azure Blob Storage enhances data integrity by ensuring records and documents are stored securely with appropriate backup measures in place.

Overall, the ARCDO Dashboard effectively modernizes the office's data management practices, improving accessibility, security, and reporting efficiency. By addressing previous



challenges associated with manual record-keeping, the platform empowers staff to manage institutional data more effectively while ensuring that critical information is easily accessible when needed. Moving forward, planned enhancements such as advanced search filters, automated MOA reminders, and improved mobile support will further extend the system's capabilities, ensuring it continues to meet the evolving needs of the institution.

VII. CONCLUSION

The development of the ARCDO Dashboard marks a significant improvement in the Alumni Relations and Career Development Office's data management capabilities. By providing a centralized, secure, and user-friendly platform for handling records related to HTEs, MOAs, OJT Coordinators, and Industry Partners, the system effectively addresses the challenges previously associated with manual documentation. The ARCDO Dashboard's features — including its interactive dashboard, expanded number cards, and streamlined export tools — are designed to enhance data accessibility and improve workflow efficiency. The implementation of the ARCDO Dashboard aligns with the institution's ongoing efforts to modernize administrative processes and improve data security. By integrating role-based access control, secure login mechanisms, and Azure-backed data storage, the system ensures sensitive information is protected while still being easily accessible to authorized users.

VIII. RECOMMENDATIONS

To further improve the ARCDO Dashboard's functionality and ensure its long-term effectiveness, the following recommendations are advised:

- **Advanced Search Filters:** Introduce search filters that allow users to refine search results by date ranges, categories, or specific keywords for faster and more accurate data retrieval.

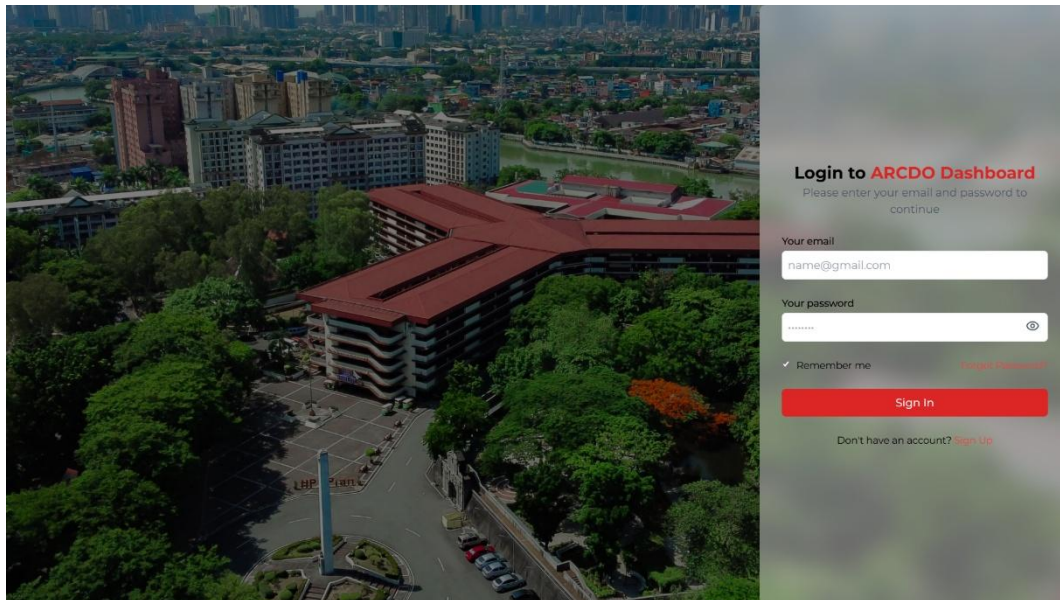


- **Automated MOA Expiration Alerts:** Implement a notification system that alerts staff about upcoming MOA expirations to ensure timely renewals.
- **Enhanced Reporting Features:** Develop customizable report templates and visual data summaries to improve documentation workflows and support strategic decision-making.
- **Mobile Optimization:** Improve the system's responsiveness for seamless access on mobile devices, allowing staff to manage data effectively while working remotely.
- **Audit Log System:** Introduce an audit log that tracks data changes, logins, and administrative actions to improve accountability and system monitoring.
- **Regular Data Audits:** Conduct routine data reviews to ensure records are accurate, consistent, and up to date.
- **User Training Programs:** Provide training sessions and user guides to improve staff understanding of system features and enhance user adoption.
- **Data Backup Strategy:** Establish a structured backup process for both the database and uploaded documents to protect against data loss.
- **System Performance Monitoring:** Regularly evaluate system performance to identify potential issues early and ensure optimal functionality.
- **User Feedback Integration:** Actively collect user feedback to identify pain points, improve features, and adapt the system to better meet institutional needs.

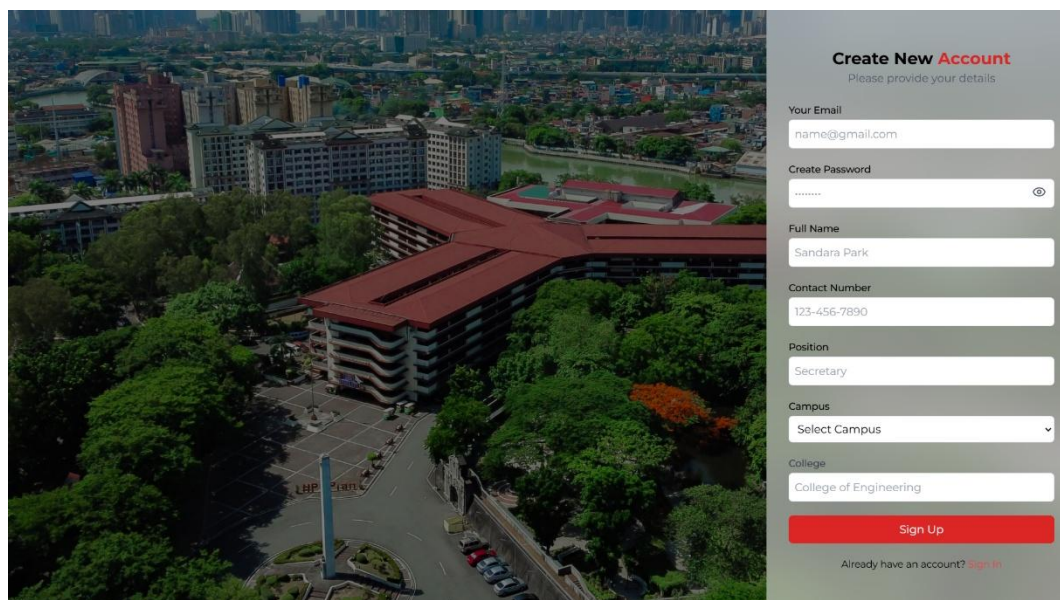
By following these recommendations, the ARCDO Dashboard will continue to evolve as a secure, efficient, and user-friendly platform that supports effective data management while addressing the institution's ongoing needs.



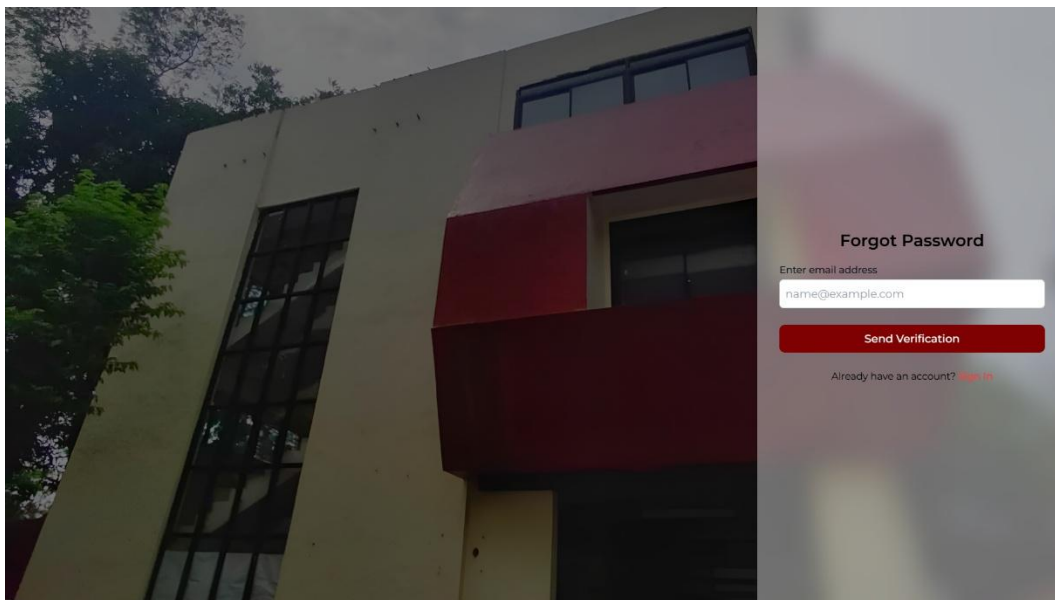
IX. ACTUAL WEBPAGE



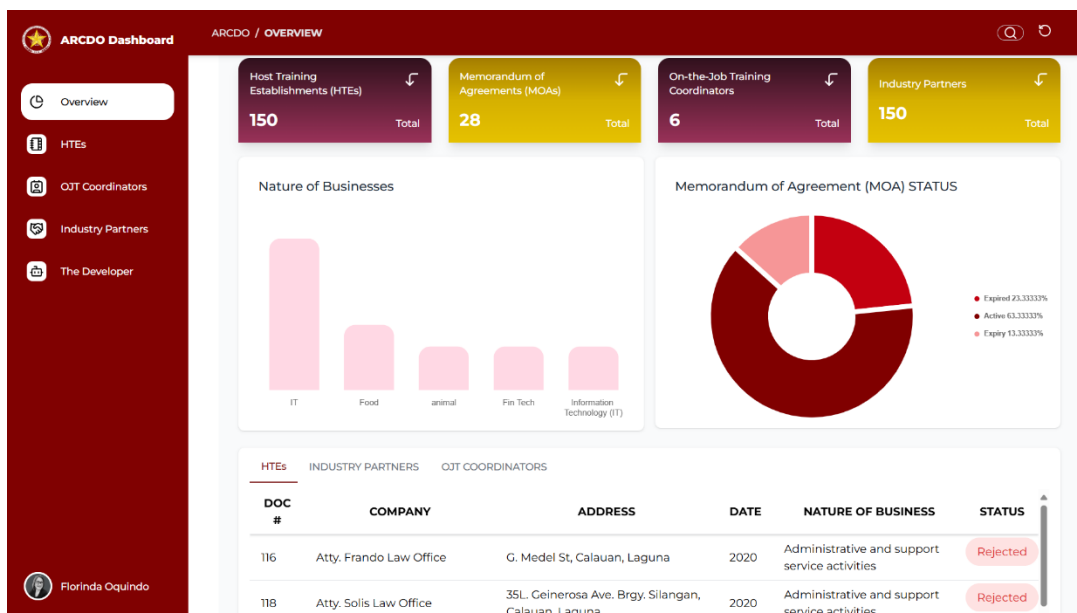
(Figure 1: Login Page)



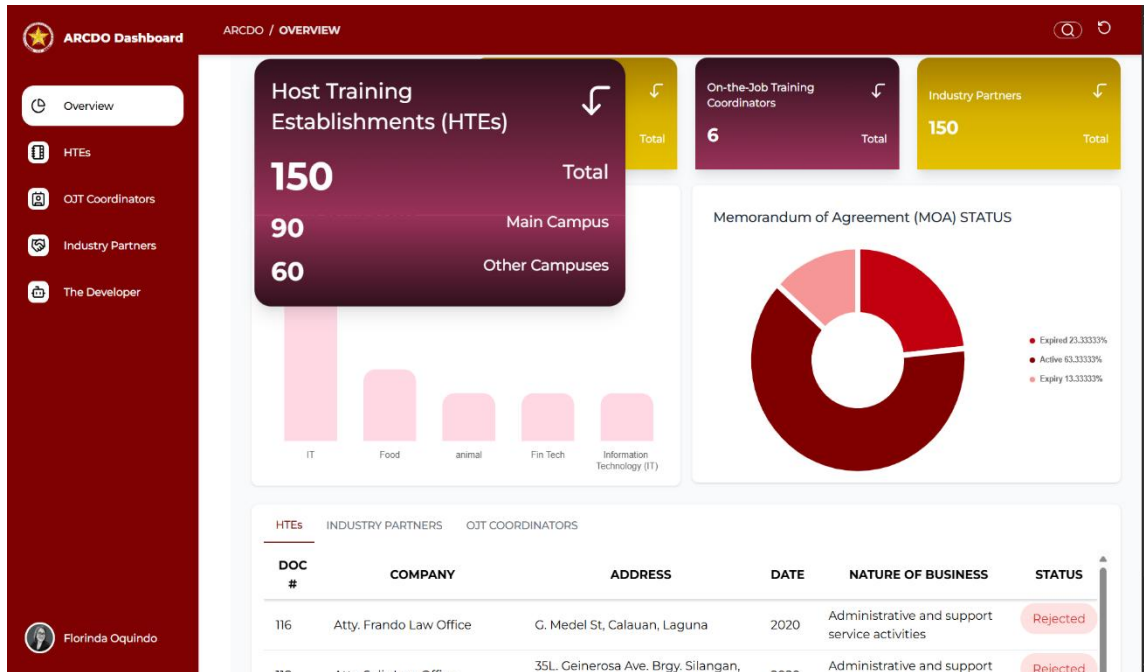
(Figure 2: Registration Page)



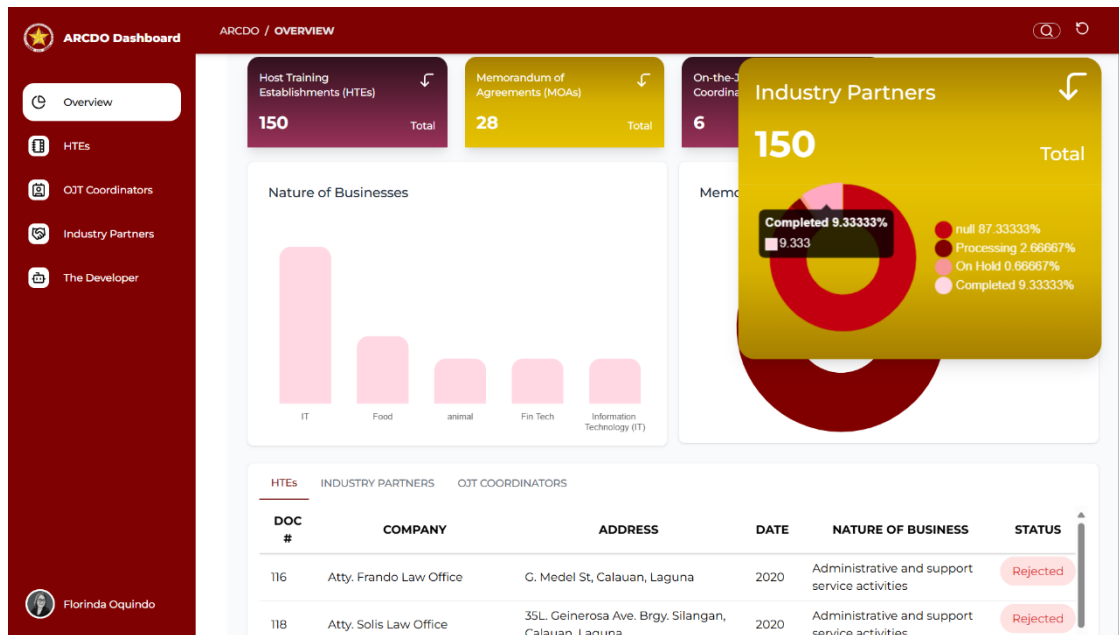
(Figure 3: Forgot Password Page)



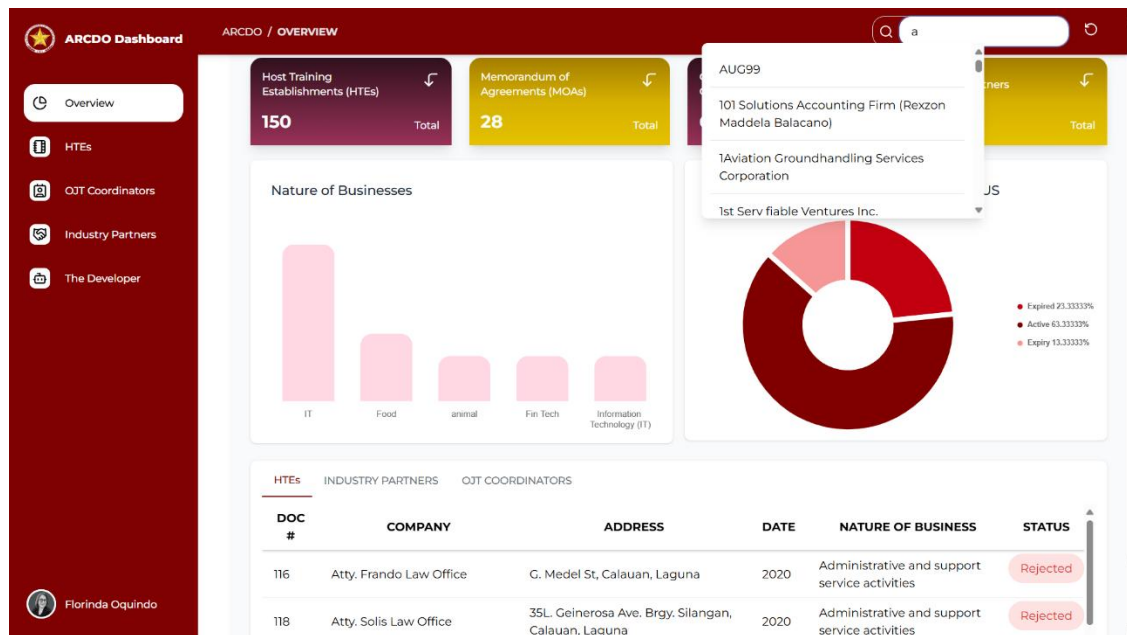
(Figure 4: Overview Page)



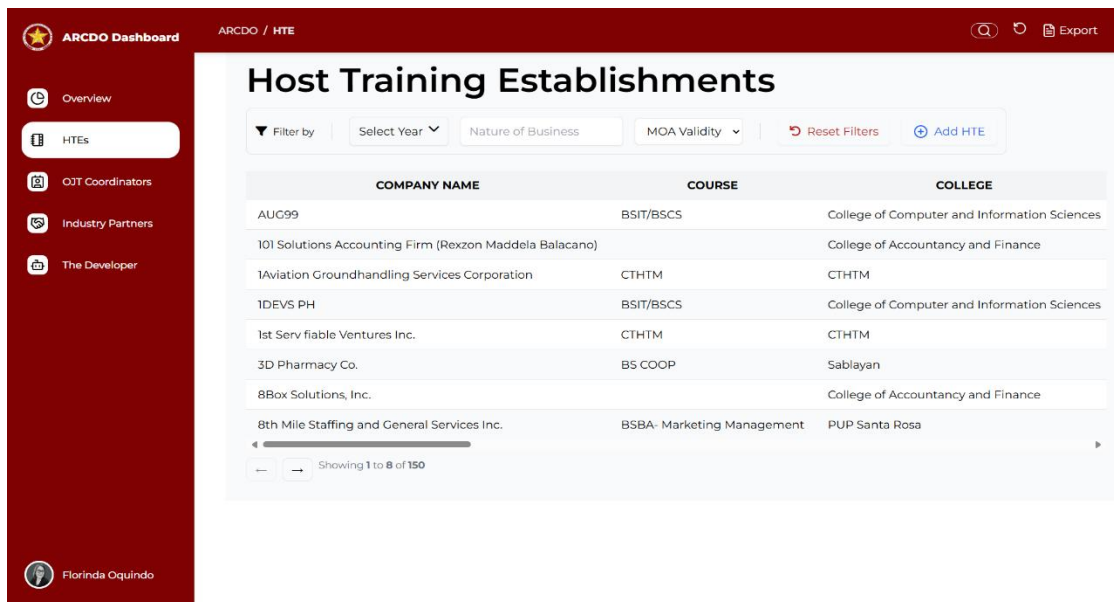
(Figure 5: Overview Page - Hovered)



(Figure 6: Overview Page - Hovered)



(Figure 7: Overview Page - Search)



(Figure 8: Host Training Establishments Page)



The screenshot shows the 'OJT Coordinators' page within the ARCDO Dashboard. The left sidebar contains navigation links: Overview, HTEs, OJT Coordinators (selected), Industry Partners, and The Developer. The main content area has a header 'OJT Coordinators' with a filter bar including 'Filter by', 'Select Campus', 'Reset Filters', and 'Add Coordinator'. Below this is a table with columns: NAME, CAMPUS, COLLEGE, EMAIL, OFFICE, ENROLLED STUDENTS, and STATUS. The table lists six coordinators with their respective details and status (On Leave, Retired, or Active). At the bottom, there are pagination controls showing 'Showing 1 to 6 of 6'.

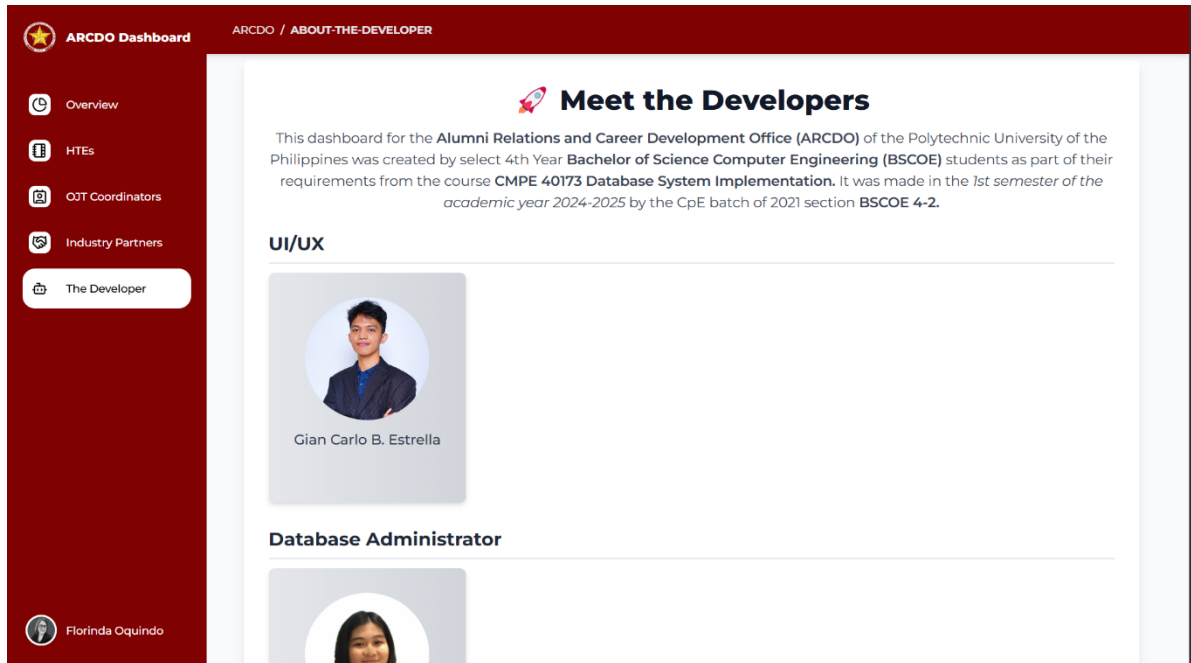
NAME	CAMPUS	COLLEGE	EMAIL	OFFICE	ENROLLED STUDENTS	STATUS
Han Soo Hee	Bataan	CCIS	rowelpogimapagmahal@gmail.com	Room 1	899	On Leave
Kim Ji Won	Main	CEA	jiwonlee@gmail.com	Room 666	2	Retired
Edwin Berico	Main	College of Engineering	edwin@gmail.com	Room 201	355	Active
Maxwell Del Valle	Quezon City	CAF	delvallemaxwell@pup.edu.ph	Room 101	50	Active
Oh Sehun	Main	CE	sehunie@pup.edu.ph	Room 212	100	Active
Jennie Kim	Quezon City	CAF	bp@gmail.com	Room 42	119	On Leave

(Figure 9: OJT Coordinators Page)

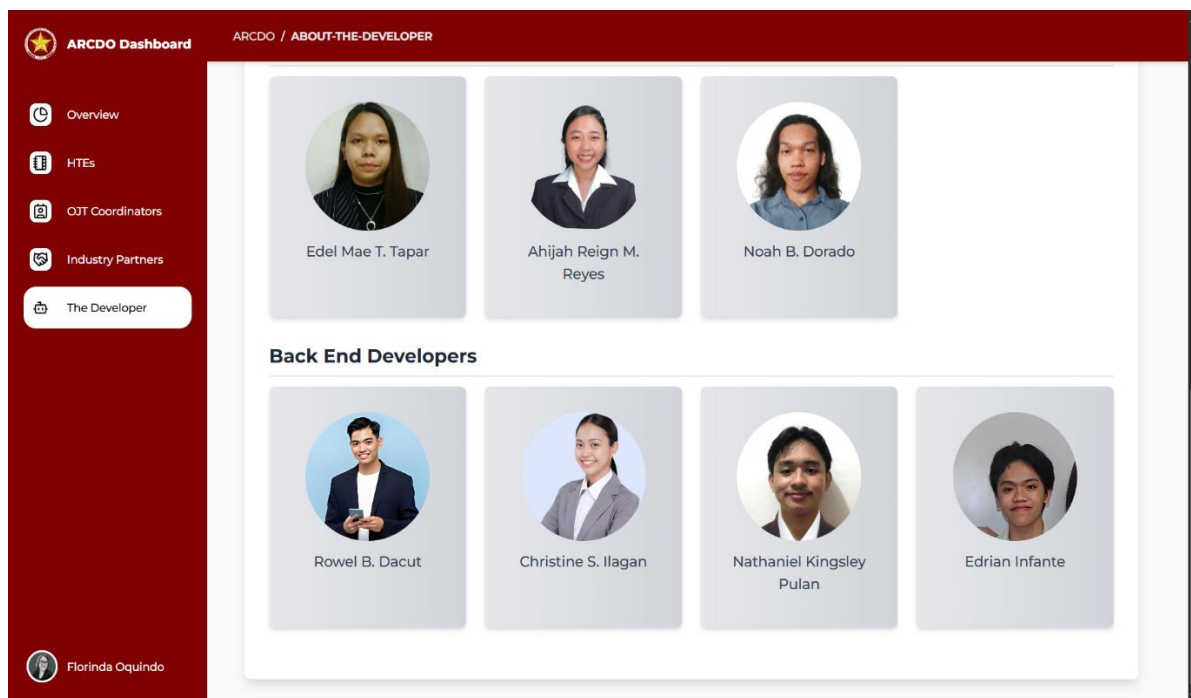
The screenshot shows the 'Industry Partners' page within the ARCDO Dashboard. The left sidebar contains navigation links: Overview, HTEs, OJT Coordinators, Industry Partners (selected), and The Developer. The main content area has a header 'Industry Partners' with a filter bar including 'Filter by', 'Select Year', 'Nature of Business', 'MOA Validity', 'Reset Filters', and 'Add Industry Partner'. Below this is a table with columns: COMPANY NAME, TELEPHONE NUMBER, FAX NUMBER, and COURSE. The table lists eight industry partners with their respective details. At the bottom, there are pagination controls showing 'Showing 1 to 8 of 150'.

COMPANY NAME	TELEPHONE NUMBER	FAX NUMBER	COURSE
1Premiere Land Marketing Co. Sta. Lucia Land Inc	9014459		Bachelor of Science in Civil Engineering
new add	12	231	
24h apartment hotel (makati)	8992424		Bachelor of Science in Hospitality Managemel
3M Service Center APAC, Inc.	79089001		BSCE
5iTalk	7966243		
7 Business Seasons Inc.	(632) 8687-5393		Bachelor of Science in Accountancy Bachelor
A. C Ong Consulting Inc.	893-5827		Juris Doctor
Abba Personnel Services, Inc.	285848215		

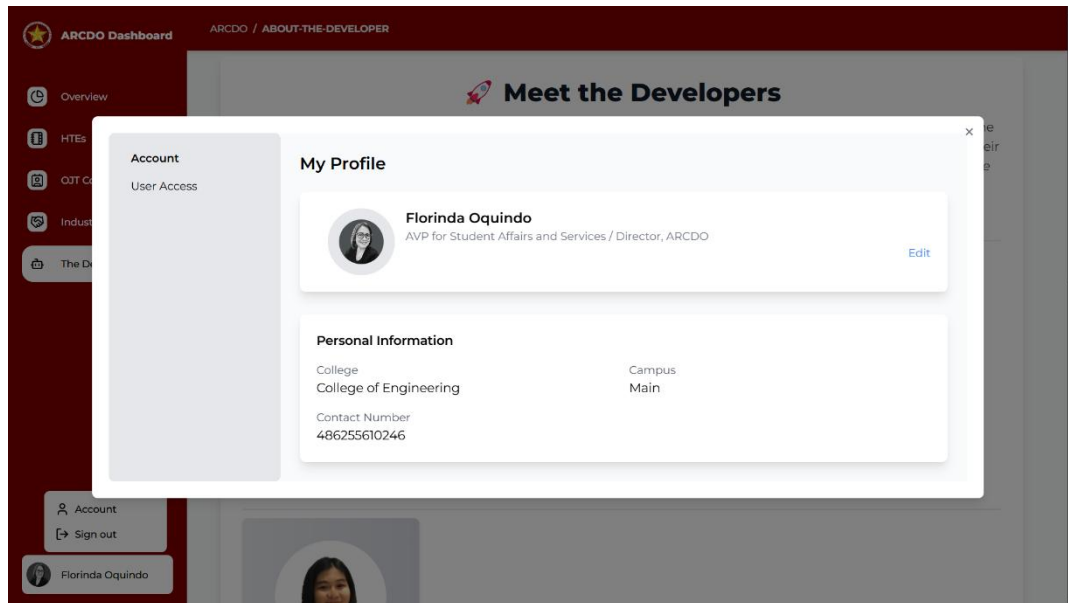
(Figure 10: Industry Partners Page)



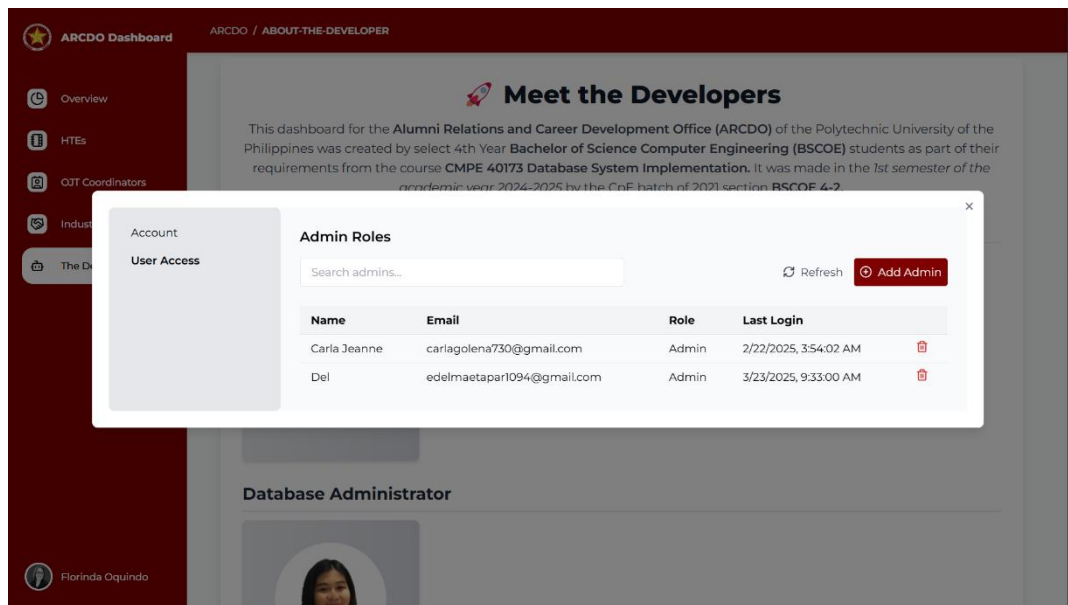
(Figure 11: About the Developers)



(Figure 12: About the Developers – cont.)



(Figure 13: Account Card)



(Figure 14: User Access Card)



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